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Sol-gel synthesis and photoluminescence characterization of $La_2Ti_2O_7:Eu^{3+}$ nanocrystals

CHENG Hua, LU Zhouguang, LIU Ying, YANG Haihua, GU Yingying, LI Wei, and TANG Yougen

College of Chemistry & Chemical Engineering, Central South University, Changsha 410083, China

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Abstract

Eu^{3+} doped $La_2Ti_2O_7$ nanocrystals with pure monoclinic phase and size of about 100 nm were prepared by a citric acid (CA) assisted sol-gel method. Techniques of thermo-gravimetric (TG) and differential scanning calorimetry (DSC), X-ray diffraction (XRD), as well as transmission electron microscopy (TEM) were employed to characterize the as-synthesized nanoparticles. Furthermore, photoluminescence (PL) performances of the Eu^{3+} doped $La_2Ti_2O_7$ nanocrystals were evaluated with focus on the effects of calcination temperature and Eu^{3+} doping concentration on the photoluminescence properties.

Keywords: lanthanum titanate; rare earth metals; photoluminescence; nanophosphors; pechini citric acid; sol-gel

1. Introduction

$A_2B_2O_7$ (A = rare earth metal, B = Ti) crystallizes in monoclinic phase with the space group of $P2_1$. The crystal structure of $La_2Ti_2O_7$ consists of layers of distorted perovskite-like slab running parallel to the {110} plane, bounded to each other by interlayer La^{3+} ions. The layers contain two different sites for La^{3+} ions in 9-fold coordination, both of which possessing C_1 point symmetry [1-3]. $La_2Ti_2O_7$ exhibits high photocatalytic activity for water splitting and decomposition of toxic compounds, such as CH_3Cl and methyl orange [4-7]. $La_2Ti_2O_7$ has found diverse applications in the electronic devices because it has the highest Curie temperature (T_c = 1500°C) so far, and interesting dielectric, piezoelectric, and ferroelectric properties [8]. $La_2Ti_2O_7$ can be used for nuclear waste disposal [9-10]. The robust layer-perovskite structure of $La_2Ti_2O_7$ shows remarkable stability against high temperature, high vacuum, as well as high-energy electrons bombarding. Both A sites and B sites in $La_2Ti_2O_7$ can be substituted by diverse component ions to form stable solid solutions. Furthermore, in the titanate-based compounds, the luminescence of active centers (Pr^{3+}, Eu^{3+}, etc.) can be excited efficiently in a wide UV range through the 4f5d band [1]. Therefore, $La_2Ti_2O_7$ is of special interest as host material for optical active centers to produce phosphors emitting a variety of colors for the development of display devices [1, 8, 11-13].

Nanoscale particle size is favorable in phosphors due to the merit of processing nanoparticles into devices with high resolution, high screen coverage, intense emission, as well as long service life [14]. The conventional synthesis method for $La_2Ti_2O_7$ is solid-state reaction. This method generally causes extensive agglomeration of the particles and compositional inhomogeneity. Therefore, several techniques were developed to synthesize ultrafine $La_2Ti_2O_7$ at moderate temperatures, such as co-precipitation [15], liquid mix techniques [16], metal organic decomposition [17], hydrothermal [6, 18], and sol-gel [5]. Sol-gel processes have attracted much attention because the intimate mixing of components in the sol-gel route ensures composition homogeneity of the final product [19]. Moreover, doping can also be easily achieved through the sol-gel protocols [20].

Here, we describe an all-aqueous sol-gel approach for the synthesis of Eu^{3+} doped $La_2Ti_2O_7$ luminescent nanocrystals. The use of citric acid (CA) as chelating agent ensures the formation of homogeneous transparent metal-citrate gels. Most importantly, the sol-gel process offers a unique, easy, stable, and low-cost sol precursor for the future fabrication of high-quality luminescent thin films by dip coating or spin coating.

2. Experimental

All the chemicals used in the experiments were of ana-

Corresponding author: LU Zhouguang **E-mail:** zglu@csu.edu.cn

lytical purity, bought from Sigma-Aldrich and used without further purification. In a typical procedure, 0.01 mol tetra-n-butyl titanate was dissolved in 50 mL anhydrous ethanol, and 1 mL concentrated nitric acid was added to restrain hydrolyzation of the tetra-n-butyl titanate. 0.02 mol citric acid was dissolved in 100 mL deionized water. Then 0.2 mol/L $La(NO_3)_3$ aqueous solution of 50 mL was added into the CA solution dropwise under vigorous stirring at 80°C to produce a clear La-citrate mixed solution. After that, the La-citrate solution was added into the tetra-n-butyl titanate ethanol solution dropwise with vigorous stirring and a transparent mixed sol was consequently produced. During this mixing procedure, the temperature was controlled at around 80°C by using a water bath. The as-obtained sol was kept at the 80°C water bath to evaporate the solvent (water and ethanol) until a transparent and viscous gel was obtained. The as-obtained gel was subsequently transferred to an oven and kept at a temperature of 160°C for 2 h to form a loose and black precursor. The precursor was then annealed at a certain temperature for 3 h with a heating rate of 10°C/min. Eu^{3+} doped $La_2Ti_2O_7$ nanocrystals were synthesized by the same route by substituting La^{3+} with a desired amount of Eu^{3+} in the raw materials preparation.

The thermal decomposition behavior of the citrate gel precursors was examined by a thermo-gravimetric analyzer (TG/DSC, Perkin Elmer, TAC 7/DX) employing air as working gas. Measurement was conducted with a heating rate of 10°C/min from 25 to 1000°C. The crystal phase of the as-prepared products was identified by powder X-ray diffraction (XRD, Bruker D8) using Cu K_α radiation (λ = 0.15418 nm) at a scan rate of 4°/min. The morphology of the final products was characterized with a Hitachi H-800 transmission electron microscope (TEM) operated at 200 kV. The high-resolution transmission electron microscopy (HRTEM) images were obtained on a JEOL-3010 equipment. The photoluminescence measurements were carried out using a Hitachi F-2500 fluorescence spectrophotometer. The excitation spectra were corrected for the beam intensity variation in the Xe light-source. For comparison, all excitation and emission spectra were measured at room temperature with constant instrumental parameters.

3. Results and discussions

3.1. Thermal analysis

The thermal decomposition behavior of the obtained La-Ti-citrate gel precursors was investigated by the TG-DSC method and the results are shown in Fig. 1. The TG curve demonstrates three continuous weight-loss regions in the range of 25-250°C (about 10%), 250-400°C (about 35%), and 400-750°C (about 12%), respectively. The first weight-loss corresponds to the elimination of physical adsorbed water. The second weight-loss region can be ascribed to the pyrolysis of NO_3^- and some organic phases (citric acid). The third weight loss is associated to the elimination of the remaining organic materials (carbon and organic compounds) and the formation of crystalline $La_2Ti_2O_7$ inorganic phase [5]. The DSC line shows four features. The endothermic process around 100°C is believed to be from the removal of the adsorbed waters. Two major exothermic peaks in the region of 350-450°C are attributed to the decomposition of the organic phase. The exothermic peak around 550°C is due to the burning out of the residual organics as well as the crystallization of $La_2Ti_2O_7$ inorganic phase, which is consistent with the TG analysis.

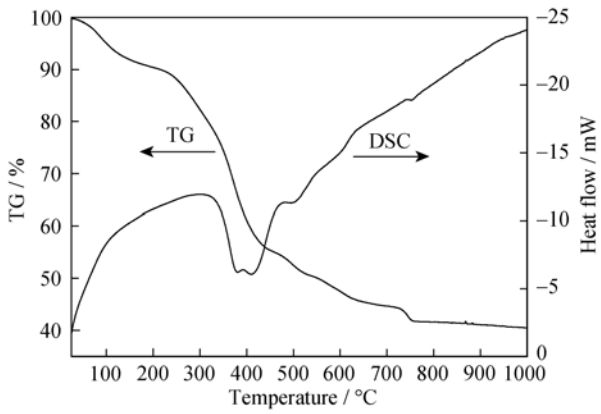

Fig. 1. TG and DSC curves of the metal-citrate gel precursors.

3.2. XRD

Fig. 2 shows the XRD patterns of the as-obtained $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanoparticles as a function of calcination temperature. The broad and weak diffraction peaks of the sample calcined at 600°C are indicative of a very fine grain size and poor crystallinity. A set of intensive diffractions can be observed from the sample calcined at 800°C. It is clear that all diffraction lines match well with the characteristic reflections of $La_2Ti_2O_7$ according to the JCPDS 28-0517 spectrum. The diffraction pattern can be indexed to the monoclinic $La_2Ti_2O_7$ with layered perovskite structure [1] and no secondary phases can be observed. When the calcination temperature was further raised to 1000°C, the integral peak intensity becomes much stronger and the full-width at half-maximum (FWHM) becomes narrower, but no secondary phase diffractions are found in the XRD patterns, revealing that the particle size increases and the crystallinity improves with temperature increasing while remaining phase pure of monoclinic $La_2Ti_2O_7$.

3.3. TEM and HRTEM

Fig. 3(a) shows the low-magnification TEM image of the

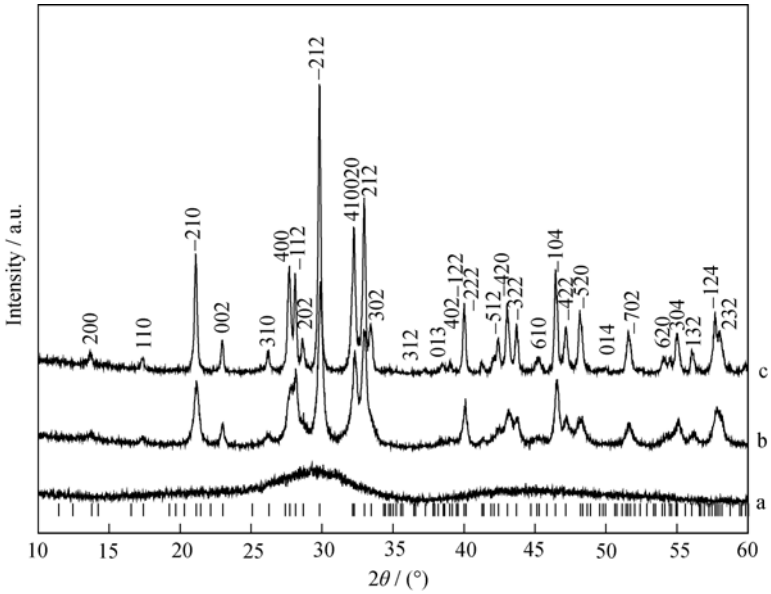

Fig. 2. XRD patterns of the products after calcining the metal-citrate gel precursors at (a) 600, (b) 800, and (c) 1000°C for 3 h in air (The composition was fixed at $(La_{0.95}Eu_{0.05})_2Ti_2O_7$, and the bottom bars indicate the observed diffractions from $Lia_2Ti_2O_7$ according to JCPDS-280517).

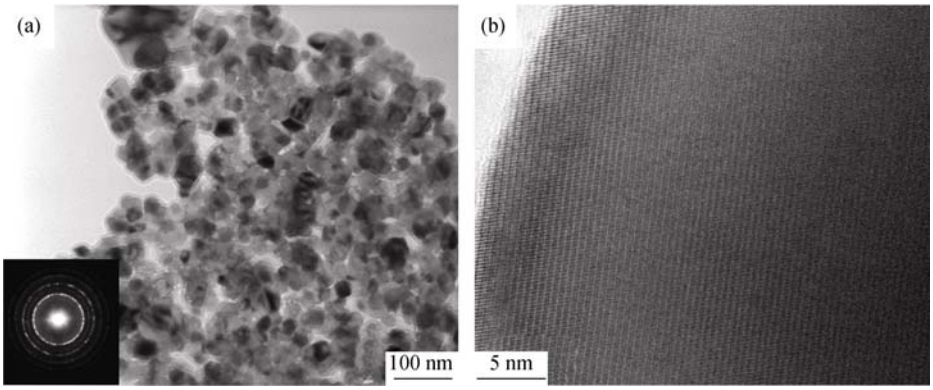

Fig. 3. (a) TEM and (b) HRTEM images of the $La_2Ti_2O_7$:Eu^{3+} (Eu:La = 0.05:0.95) nanoparticles calcined at 800°C for 3 h (The inset in (a) is the corresponding selected area electron diffraction (SAED) pattern).

as-obtained $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanoparticles. It reveals that the sample consists of uniform nanoparticles with a grain size of about 60 nm. The SAED pattern demonstrates well-defined diffraction rings, indicating good crystallinity and poly-crystal nature of the synthesized $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanoparticles. The order lattice fringes of the HRTEM images as revealed in Fig. 3(b) further confirms that the $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanoparticles are well-crystallized. It also suggests that the Eu^{3+} active ions have been effectively doped into the crystal lattice of the $La_2Ti_2O_7$ nanoparticles.

3.4. Photoluminescence

Fig. 4 shows the room-temperature excitation and emission spectra of the $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanophosphors calcined at 800°C. The broad band between 220 and 300 nm in the excitation spectrum (Fig. 4, left) can be attributed to the charge transfer band (CTB) between the Eu^{3+} and the surrounding oxygen anions [21-23]. The emission spectrum (on the right of Fig. 4,) mainly appears in the red spectral area (from 550 to 710 nm), corresponding to transitions from the excited $^5D_0$ state to the ^7F_J (J = 0, 1, 2, 3, 4) levels of the $4f^6$ configuration of Eu^{3+} [21]. As shown in Fig. 4, the strongest emission band for the as-obtained $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanophosphors is centered at around 614 nm originating from the electric dipole transition $^5D_0$-$^7F_2$. While the emission band centered at around 588 and 596 nm, originating from the $^5D_0$-$^7F_1$ magnetic dipole transition, is much weaker. Theoretically, when the Eu^{3+} ions are embedded in a site of non-inversion symmetry, the $^5D_0$-$^7F_2$ electric dipole transition is dominant, while in a site with inversion symmetry, the $^5D_0$-$^7F_1$ magnetic dipole transition is the strongest [21, 24-25]. As a result, the photoluminescence spectrum reveals

a non-inversion symmetry environment for the Eu^{3+} active centers in the $La_2Ti_2O_7$ crystal structure. From a crystallographic point of view, the crystal structure of $La_2Ti_2O_7$ consists of distorted perovskite-like slabs running parallel to the {110} plane, bounded to each other by interlayer La^{3+} ions [1, 8]. The site symmetry of the La^{3+} site is noncentrosymmetric C_1 in the monoclinic $P2_1$ lattice of $La_2Ti_2O_7$ at room temperature [8]. Therefore, the Eu^{3+} that substitutes the La^{3+} should also locate in an asymmetric site, which is consistent with the PL results.

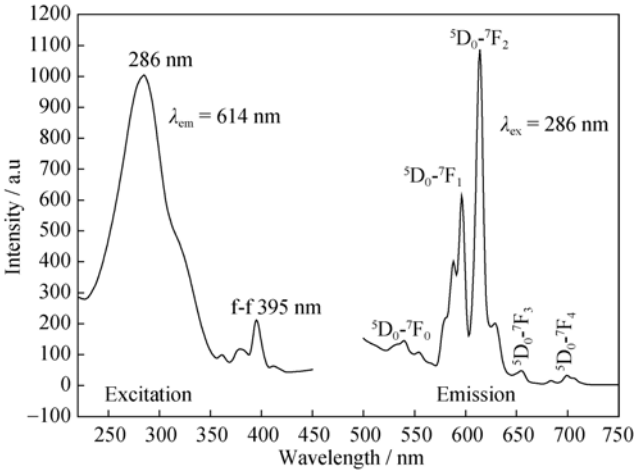

Fig. 4. Typical room temperature excitation and emission spectra of the $La_2Ti_2O_7:Eu^{3+}$ (Eu:La = 0.05:0.95) nanophosphors calcined at 800°C for 3 h.

Fig. 5 summarizes the emission intensity of the $(La_{0.95}Eu_{0.05})_2Ti_2O_7$ nanocrystals as a function of the calcination temperature. The emission intensity increases with the calcination temperature from 500 to 1000°C. The weak emission at lower temperature is presumably due to the disorder crystal structure and the remaining quenching centers such as impure NO_3^- and OH^- ions. When the calcination temperature increases, the crystallinity improves and the quenching centers largely reduce, resulting in a brighter emission. [25-26]

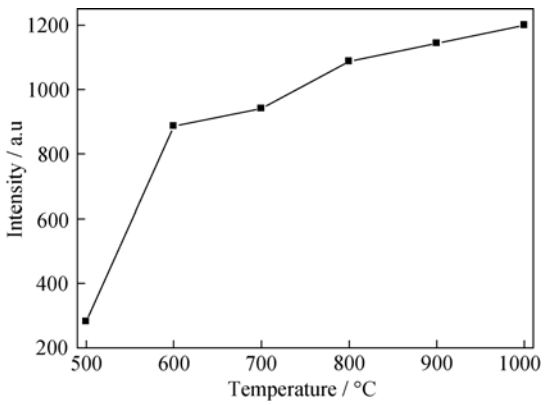

Fig. 5. The emission intensity (614 nm) resulting from the $^5D_0$-$^7F_2$ transition as a function of the calcination temperature (The composition was fixed at $(La_{0.95}Eu_{0.05})_2Ti_2O_7$).

The effects of the Eu^{3+} active center concentration on the emission intensity of the as-synthesized $(La_{1-x}Eu_x)_2Ti_2O_7$ nanocrystals are shown in Fig. 6. It reveals that the emission intensity increases with the Eu^{3+} concentration, and reaches the strongest emission when the Eu^{3+} concentration approaches around 5%, thereafter, the luminescence tends to decrease slightly. The PL intensity is highly dependent on the activator (Eu^{3+}) concentration. Generally speaking, the intensity increases with activator concentration up to a certain extent. Above this point, pairing or aggregation of activator atoms leads to efficient resonant energy transfer between Eu^{3+} ions and a fraction of energy migration to distant killer or quenchers, and quenching behavior appears, resulting in reduced PL intensity [27]. It is evident that for this type of phosphors, quenching starts at a Eu^{3+} atomic concentration of about 5%.

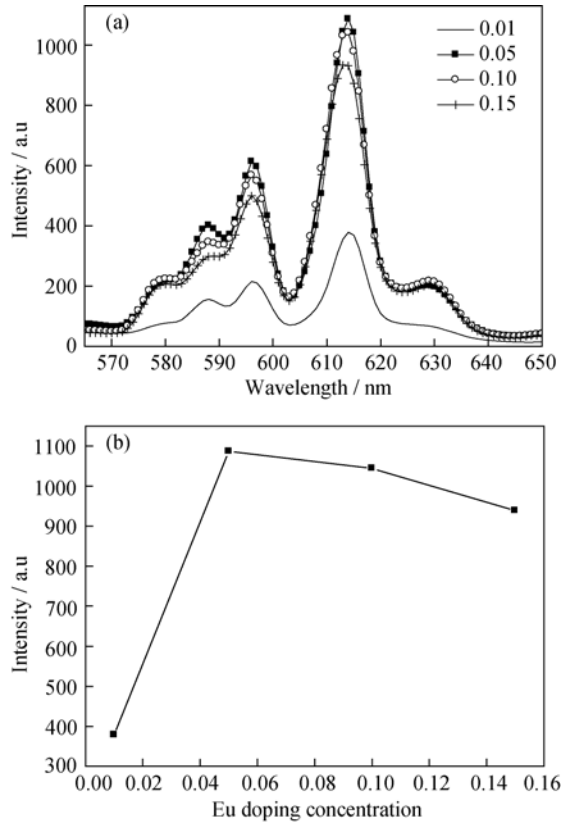

Fig. 6. (a) evolution of the emission spectra of the $La_2Ti_2O_7:Eu^{3+}$ nanophosphors calcined at 800°C for 3 h with different Eu^{3+} doping concentrations, and (b) the emission intensity of the peak centered at 614 nm ($^5D_0$-$^7F_2$ transition) plotted as a function of Eu^{3+} active center concentration.

4. Conclusions

In summary, Eu^{3+} doped $La_2Ti_2O_7$ nanocrystals have

been successfully synthesized via a facile citric acid-assisted sol-gel approach. Single monoclinic phase $La_2Ti_2O_7$ nanoparticles with perovskite structure and the particle size of 60 nm can be reproducibly obtained at 800°C. PL characterization shows that after doping Eu^{3+} in the La^{3+} sites, the as-synthesized $La_2Ti_2O_7$ nanocrystals display dominant emissions in the red area (from 550 to 710 nm) with the strongest line at 614 nm, corresponding to the $^5D_0$-$^7F_2$ electric dipole transition of the Eu^{3+}. The optimized doping concentration of Eu^{3+} was found to be around 5 at.%.

Acknowledgements

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